SCIENCE FOUNDATIONS

Experimental Error

How good is my data?

PART 1

Where Does Error Come From?

Why We Repeat Experiments

↕ Some Trials Are High, Some Are Low

One measurement is a guess. Repeat it, and some trials come out a bit high, others a bit low.

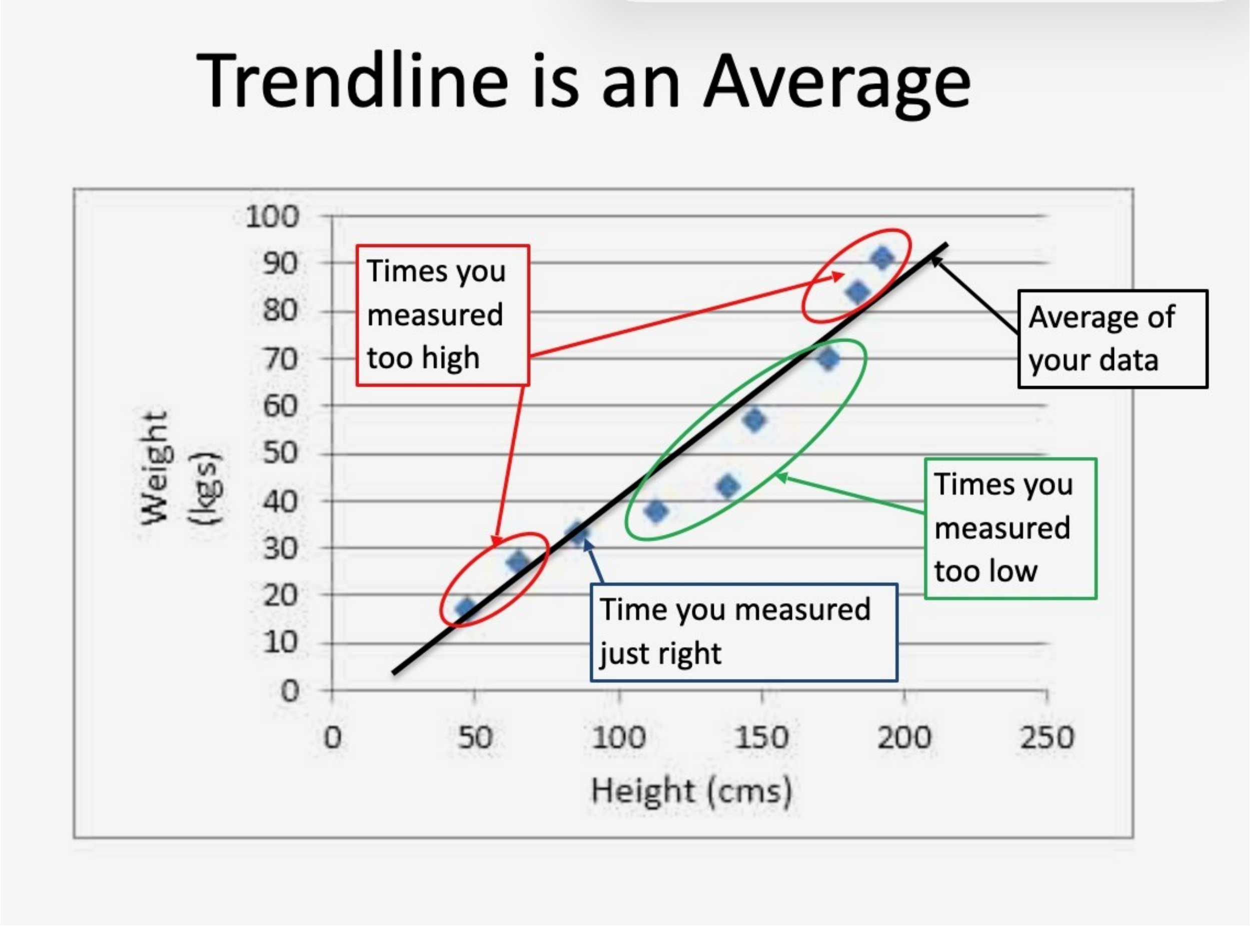
The HIGHS and LOWS cancel each other out when you AVERAGE them — which brings you close to the true value.

~ The Trendline IS the Average

When you plot your repeated trials on a graph, the trendline shows the average pattern — it's what you'd expect if random error had cancelled itself out.

More trials → tighter average → better estimate of truth.

The Trendline IS the Average



WHERE EVERY MISTAKE COMES FROM

The Five Sources of Error

1 **YOU**

Reaction time, eyesight, technique, mood.

2 **Your Tools**

Every ruler, scale, or timer has a limit.

3 **Manipulation**

Nudging or moving the setup while measuring.

4 **External influences**

A/C, drafts, vibrations, temperature drift.

5 **Experimental design**

Poor controls or too many variables changing.

PART 2

Precision vs. Accuracy

THEY MEAN VERY DIFFERENT THINGS

Two Words That Sound Alike

P

PRECISION

How close your repeated measurements are to EACH OTHER.

Repeatability.

You can spot precision problems DURING the experiment — if one trial is way off, just repeat it.

A

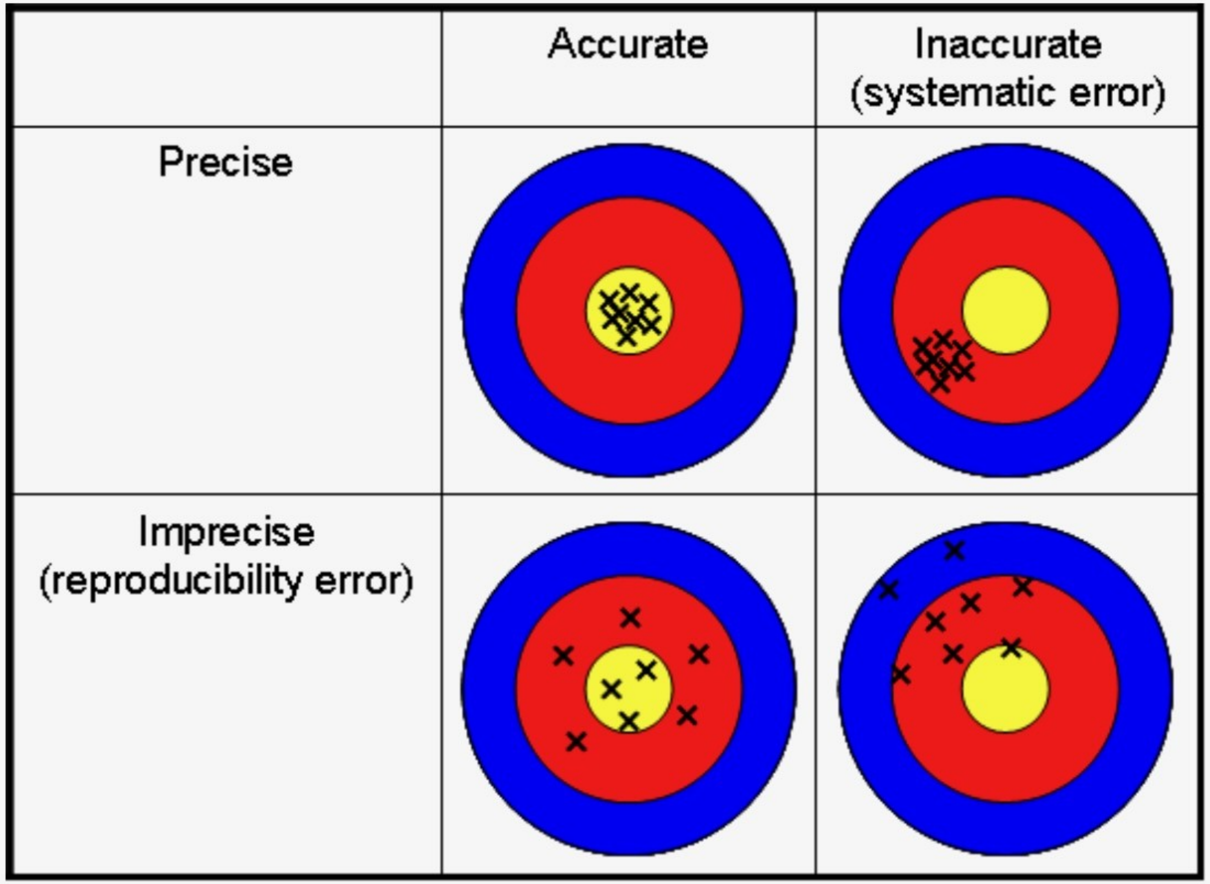
ACCURACY

How close your data is to the EXPECTED (true) value.

Hit the target.

You can only calculate accuracy AFTER the experiment — using percent error.

The Bullseye Test



Top-Left ✓✓

Precise AND accurate — the goal.

Top-Right

Precise but off-target — systematic error (bad calibration).

Bottom-Left

Accurate on average, but noisy — reproducibility error.

Bottom-Right ✗

Neither — go home and start over.

WATCH DATA AS IT COMES IN

Precision — Spot the Outlier

Time for a runner to finish a race (seconds):

TRIAL 1	TRIAL 2	TRIAL 3	TRIAL 4
12 s	27 s	13 s	11 s
consistent	*OUTLIER*	*consistent*	*consistent*

The 27 s is obviously wrong. Repeat that trial — don't average bad data in.

Technique Matters Too

Measuring room length with meter sticks — which is more precise?

✗ Sticks END-to-END IN THE AIR

Wobble. Shake. Gaps. Bad joints.
Small errors on every seam add up.

✓ Sticks END-to-END ON THE FLOOR

Sticks stay flat. Seams are tight.
Repeatable — much more precise.

PART 3

Error & Percent Error

The Error Formula

$$\text{Error} = |\,\text{accepted} - \text{experimental}\,|$$

|

Why Absolute Value?

The vertical bars | | make the result POSITIVE, whether you measured too high or too low.

Error just tells us HOW FAR OFF — never a negative distance.

kg

Same Units As Original

Error is in the SAME units as your measurement.

Measured 175,000 kg vs. accepted 180,000 kg → Error = 5,000 kg.

Why Isn't Error Enough?

A Blue Whale

Accepted mass: 180,000 kg

You measure: 175,000 kg

Error = 5,000 kg

SOUNDS HUGE!

% Error = 2.8%

A House Cat

Accepted mass: 8 kg

You measure: 7 kg

Error = 1 kg

SOUNDS TINY!

% Error = 12.5%

The Percent Error Formula

% Error = (Error ÷ Accepted) × 100%

The closer your percent error is to 0, the more ACCURATE your data.

THEY ALL MEAN THE SAME THING

You'll See This Formula In Different Ways

MOST TEXTBOOKS	SCIENCE JOURNALS	LAB REPORTS
$$\frac{\lvert \text{accepted} - \text{experimental} \rvert}{\text{accepted}} \times 100\%$$	$$\frac{\lvert \text{observed} - \text{true} \rvert}{\text{true}} \times 100\%$$	$$\frac{\lvert \text{measured} - \text{actual} \rvert}{\text{actual}} \times 100\%$$
Result: no units, just add a %	*Result: no units, just add a %*	*Result: no units, just add a %*

Worked Example — Density of Water

You measure water's density as 0.96 g/mL. The accepted value is 1.00 g/mL. Find the error and percent error.

1

IDENTIFY

Accepted = 1.00 g/mL
Experimental = 0.96 g/mL

2

ERROR

|1.00 − 0.96|
= 0.04 g/mL

3

% ERROR

(0.04 ÷ 1.00) × 100%
= 4%

4

JUDGE

4% error — pretty good for a middle-school lab.

Practice — Counting Raisins

1

ESTIMATE

How many raisins do you think are in the box?

2

COUNT

Pour them out. Count the actual number.

3

FIND ERROR

Error = |actual − estimate|

4

FIND % ERROR

% Error = (Error ÷ actual) × 100%

HOMEWORK

Experimental Error Problem Set #1

Posted on Google Classroom. Due next class meeting.

Remember: identify accepted, experimental, error, and % error — with units.